

Chapter-6

"Color Image Processing"

Color spectrum of light is divided into 6 regions :

Violet - Blue - Green - Yellow - Orange - Red

No color in the spectrum ends abruptly but rather bends smoothly into the next.

The electromagnetic spectrum is split up according to the wavelengths of different forms of energy. ^{only} light is sensed by the human eye.

A body that reflects light, that is balanced in all visible wavelengths appears white.

The human eye combines 3 primary colors to discern all possible colors.

NOTE

Red $\rightarrow$ 700 nm wavelength.

Green $\rightarrow$ 546.1 nm wavelength.

Blue $\rightarrow$ 435.8 nm wavelength.

Retina contains 2 types of photoreceptors

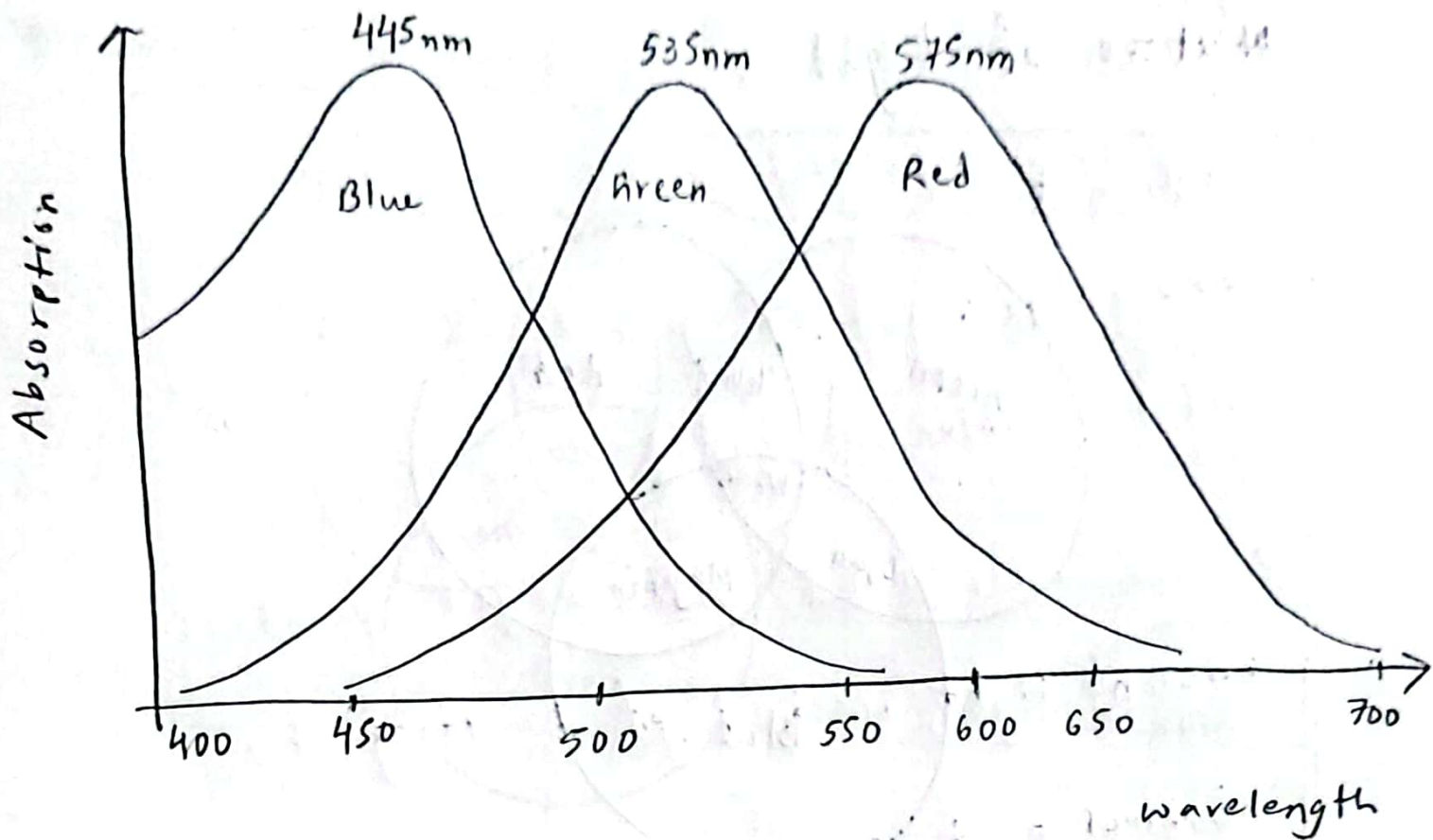
- (i) Cones $\rightarrow$ color sensitive.
- (ii) Rods $\rightarrow$ insensitive to color.

6-7 millions cones are present in the eye.

65% is sensitive to red light.

33% is sensitive to green light.

2% is sensitive to blue light.



Primary colors of light are additive

Primary colors are red, green & blue.

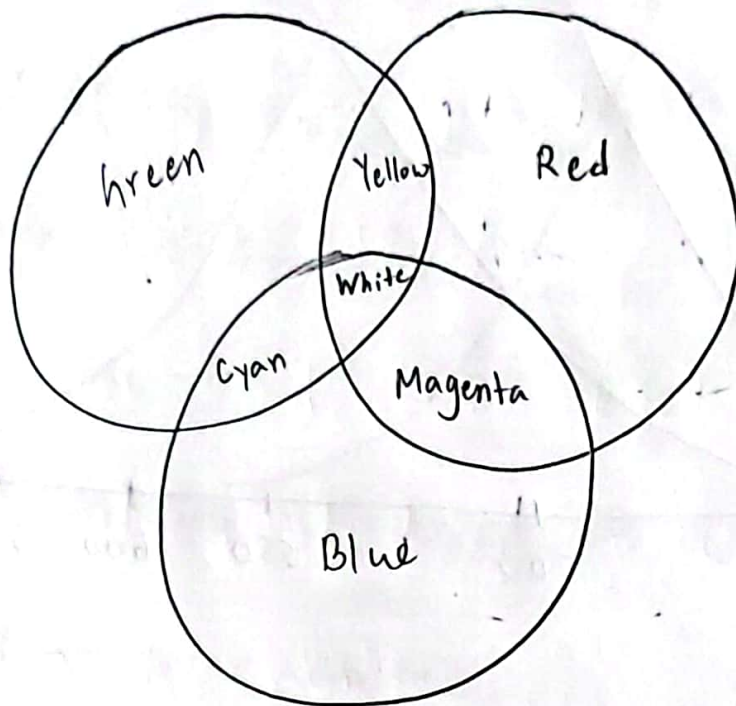
red + green + blue $\rightarrow$ white

Primary colors of pigments are subtractive

Primary colors are cyan, magenta & yellow.

cyan + magenta + yellow $\rightarrow$ black.

Mixture of Light



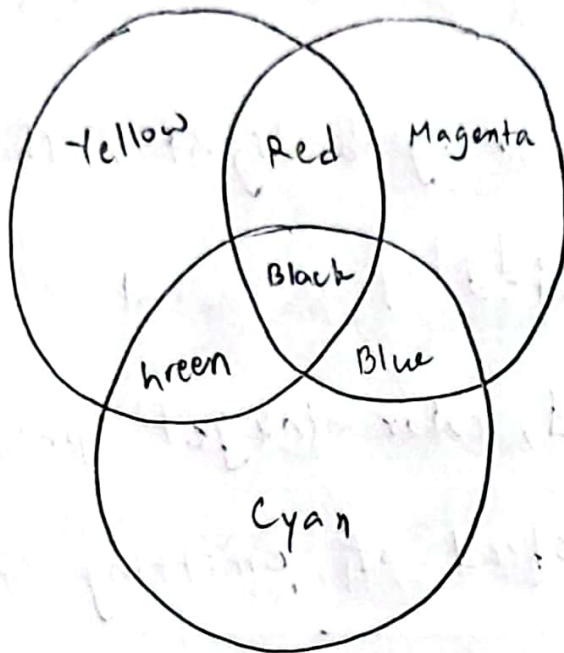
Green + Red $\rightarrow$ Yellow

Green + Blue $\rightarrow$ Cyan

Red + Blue $\rightarrow$ Magenta

R + G + B $\rightarrow$ White

Mixture of pigments



Yellow + Magenta $\rightarrow$ Red

Cyan + Yellow $\rightarrow$ Green

Magenta + Cyan $\rightarrow$ Blue

$C + M + Y \rightarrow$ Black

Magenta = White - Green

Cyan = White - Red

Yellow = White - Blue

NOTE

Computer monitors & TV sets emit combinations of r g b light. This is an additive color model.

On the other hand, color inkjet printers, absorb light instead of emitting it.

Combination of cyan, magenta and yellow inks are used. This is a subtractive color model.

NOTE

R + G + B colors do not produce white when we mix them. This is because we mix pigments.

To make red paint, we mix red pigments - small particles that reflect red light and absorb green and blue light.

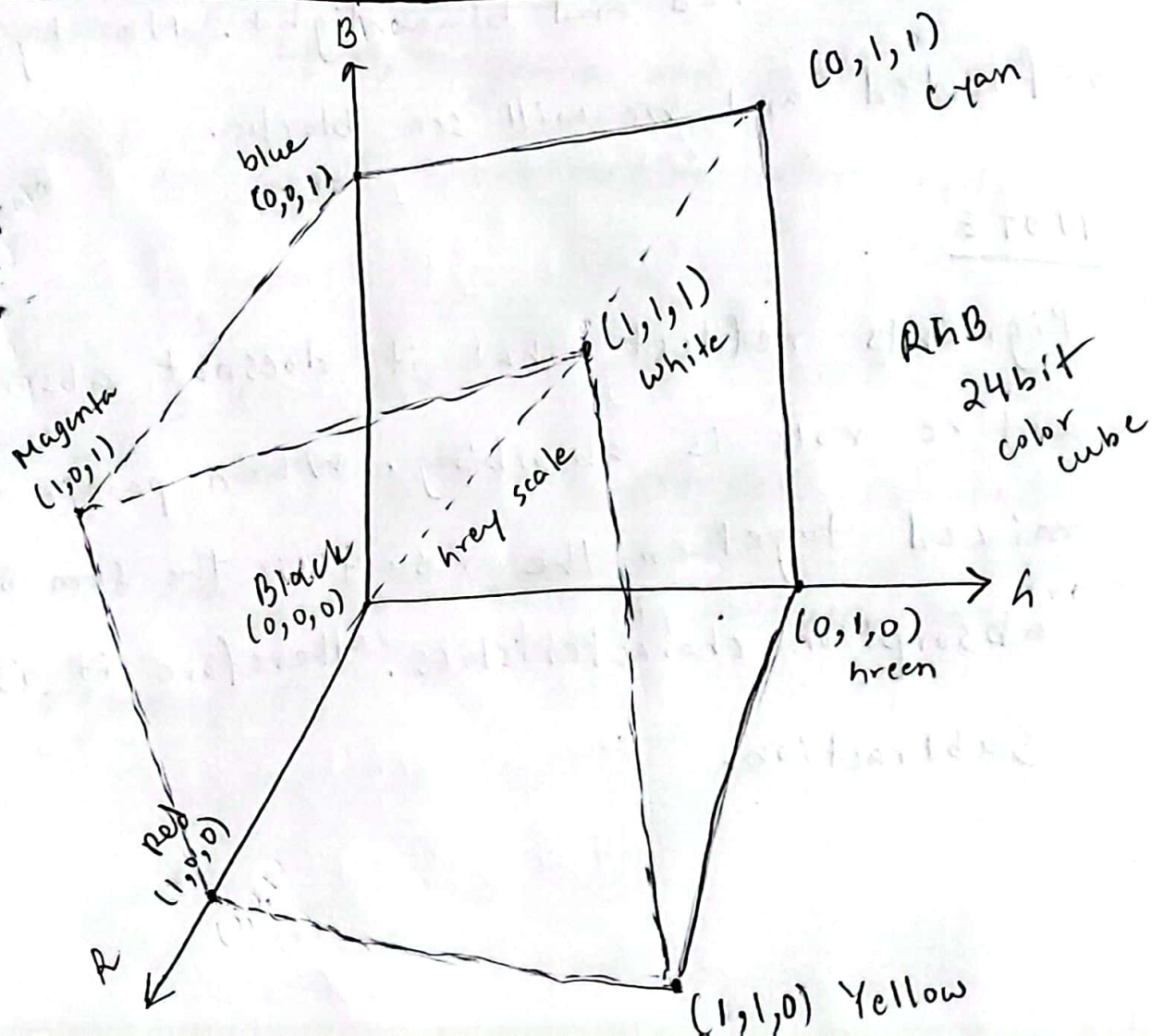
If we mix red and green pigment in white paint, the red pigment will absorb the green and blue light while the green pigment will absorb the red and blue light. No light is reflected and we will see black.

NOTE

Pigments reflect what it doesnot absorb but its active role is absorbing. When paints are mixed together, the result is the sum of their absorption characteristics. Therefore it is color subtraction.

Pigments in filters selectively absorb different colors and the remaining colors pass through the filters just as the remaining colors are reflected from paint.

Cartesian co ordinate system



$(2^8)^3$ colors possible .

Color quantization

24 bits $\rightarrow$ 8 bits

then 3 bits for red

3 bits for green

2 bits for blue

$$\text{intensity} = 0.299 \times \text{red} + 0.587 \times \text{green} + 0.144 \times \text{blue} .$$

$\downarrow$
appears
brightest

if 3 light sources have the same actual intensity but are colored red, green and blue, green appears the brightest followed by red and blue .

————— END of chapter - 6 —————